Deepak Gopinath

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Work Experience

Thinking Machines Lab — New York City, NY / San Francisco, CA
Member of Technical Staff, currently among the 40 longest tenured employees — Oct 2025 – current

- Developed multimodal pre-training and post-training data, especially to address split brain model behavior, while owning data and inference infra to produce datasets at scale, and repeat-aware pretraining scaling ladder to derisk datasets; and long horizon agentic post-training and evaluation, for the release of Interaction and Inkling Models.
- Co-developed pre-training and post-training evaluation infrastructure – setting up core evaluation suites and debugging evaluation correctness across verticals; reliable, fault-tolerant and throughput-optimized inference; and an interactive dashboard to visualize and compare model/grader rollouts, metrics and configurations.
- Skills: Python, PyTorch, Ray Core, Spark, Kafka, SGLang, Databases

Apple Foundation Models — New York City, NY
Machine Learning Researcher, LLMs — March 2023 – Oct 2025

- Large Language Model (LLM) research as part of the core model training team for Foundation Models (FM) that power Apple Intelligence and FM Developer API [blog][technical report]
- Led efforts on pre-training and evaluation of language models, with a focus on web scale data, pretraining recipes and multilinguality, with multiple work streams that collectively engaged 25+ contributors
- Skills: Python, Jax, Apache Beam on Dataflow, AXLearn/Ajax LLM training frameworks

Meta AI (Facebook AI Research) — Menlo Park, CA / Pittsburgh, PA
Research Engineer, AI for Creativity — July 2019 – March 2023

- Lead Engineer for AI for Creativity supporting exploratory research in animating human characters with transformers and RL. Developed and open-sourced the fairmotion and ScaDiver libraries [12]
- Led cross-functional projects to apply research to production. Established collaborations with Habitat for Embodied AI, Meta Horizon Worlds, Sanzaru game studio, FAIR Paris/Menlo Park, and Movement Lab at Stanford.
- Published research at ACM SIGGRAPH [7] [10] [11], NeurIPS [9], and SIGGRAPH Asia [8]
- Tech lead for multiple projects involving 20+ engineers in total at Meta AI. Mentored 9 Research and SWE interns, supported 20+ new team members and mentored 5 junior engineers in technical mentorship programs
- Skills: Python, PyTorch, TorchServe; RLLib for distributed RL training; PyBullet and Isaac Gym physics simulators; OpenGL, Blender, Unity for character/scene building, animation and rendering.

Research Engineer, Language and Translation Technologies — Jan 2017 – July 2019

- Co-developed and shipped the first neural machine translation (NMT) system to power translations for all user content on Facebook and Instagram (7B+ translations per day) [blog][press]
- Tech lead for a team that modeled and deployed the first multilingual NMT systems to production [blog] [press]
- Managed "engineering excellence" projects towards optimized workflows for data processing, model training and evaluation, transitioning models to PyTorch, addressing reproducibility, determinism and safety in translations.
- Skills: Python, PyTorch; ONNX, C++ for production code; Hive, Spark for data processing

Software Engineering Intern, Language Technology — May 2016 – Aug 2016

- Developed neural network language models on the (newly) open sourced deep learning framework Caffe2.0

Education

Carnegie Mellon University — Pittsburgh, PA
M.S., Language Technologies Institute, School of Computer Science — Aug 2015 – Dec 2016

Birla Institute of Technology and Science, Pilani — Pilani, India
B.S. in Computer Science — 2010 – 2014

Early Career

LinkedIn Bangalore, India
Software Engineer, Content Search July 2014 – Aug 2015

- Developed the prefix query tagger library to predict autocompletions and tags for an incomplete query
- Co-implemented and deployed federated search and auto-complete search for several verticals on linkedin.com
- Actively involved in recruitment and onboarding of university grad hires and co-mentored 3 SWE interns
- Skills: Java, Lucene, Pig, Flex, Play Framework with JQuery and Bootstrap, JavaScript

Microsoft Research Bangalore, India
Research Developer Intern, Applied Sciences Jan 2014 – July 2014

- Fraud detection in Azure services and sentiment analysis of Azure users in social media
- Launched the Indian Elections Game web app, an interactive prediction market for the public to make predictions on the 2014 Indian Elections [blog]

Amazon Bangalore, India
Machine Learning Intern May 2013 – July 2013

- Adult content detection for Amazon Android AppStore, leading to a 30% load reduction for human reviewers
- Developed a system to cluster Amazon seller FAQs using Latent Dirichlet Allocation (LDA) and map customer queries to the clusters with logistic regression, SVMs, and supervised LDA

Awards and Honors

- ACM ICMI Ten-Year Technical Impact Award Runner-Up Award for the paper "Deep Multimodal Fusion for Persuasiveness Prediction" to be awarded at ICMI 2026 in Napoli, Italy
- Project Acyut: A team of 5 undergraduate students that built India's first indigenously developed autonomous humanoid robot. I designed and implemented all aspects of the computer vision and robot behavior modules [press coverage]
  - 3rd Prize at RoboCup 2011 in Autonomous Humanoid Team Soccer
  - 2 silver medals at RoboGames 2011, the (then) world's biggest open robotics event
- Placed 4th in ACM ICPC, Asia Regionals, Kharagpur site in December 2013
- Placed 1st in LinkedIn Bangalore HackDay 2015 and 2nd in Amazon Machine Learning HackEnd 2013

Publications

[1] "Over-Searching in Search-Augmented Large Language Models" [pdf]
R. Xie, **D. Gopinath**, D. Qiu, D. Lin, H. Sun, S. Potdar, and B. Dhingra
19th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026.

[2] "Interleaved Reasoning for Large Language Models via Reinforcement Learning" [pdf]
R. Xie, D. Qiu, **D. Gopinath**, D. Lin, Y. Sun, C. Wang, S. Potdar, B. Dhingra
NeurIPS Workshop on Multi-Turn Interactions in Large Language Models (MTI-LLM), 2025.

[3] "Apple Intelligence Foundation Language Models" [pdf]
T Gunter, Z Wang, C Wang, R Pang, ..., **D. Gopinath**, Apple Inc.
arXiv preprint arXiv:2407.21075, 2024.

[4] "Simulation and Retargeting of Complex Multi-Character Interactions" [pdf]
Y Zhang, **D. Gopinath**, Y Ye, J Hodgins, G Turk, J Won
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2023.

[5] "CIRCLE: Capture In Rich Contextual Environments" [pdf]
JP Araújo, J Li, K Vetrivel, R Agarwal, J Wu, **D. Gopinath**, AW Clegg, K Liu
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023.

[6] "Motion In-betweening for Physically Simulated Characters" [pdf]
D. Gopinath, J. Won, and H. Joo
ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia), 2022.

[7] "Physics-based Character Controllers using Conditional VAEs" [webpage][pdf]
J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2022.

[8] "Transformer Inertial Poser: Attention-based Real-time Human Motion Reconstruction from Sparse IMUs" [pdf]
Y. Jiang, Y. Ye, **D. Gopinath**, J. Won, A.W. Winkler, and C.K. Liu
ACM Special Interest Group on Computer Graphics Asia (SIGGRAPH Asia), 2022.

[9] "Leveraging Demonstrations with Latent Space Priors"
J. Gehring, **D. Gopinath**, J. Won, G. Synnaeve, A. Krause, and N. Usunier
Conference on Neural Information Processing Systems (NeurIPS) (under review), 2022.

[10] "Control Strategies for Physically Simulated Characters Performing Two-player Competitive Sports". [webpage][pdf]
J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2021.

[11] "A Scalable Approach to Control Diverse Behaviors for Physically Simulated Characters" [webpage][pdf]
J. Won, **D. Gopinath**, and J. Hodgins
ACM Special Interest Group on Computer Graphics (SIGGRAPH), 2020.

[12] "fairmotion - Tools to load, process and visualize motion capture data" [code]
D. Gopinath and J. Won
GitHub, facebookresearch/fairmotion, 2020.

[13] "Harnessing Indirect Training Data for End-to-End Automatic Speech Translation: Tricks of the Trade" [pdf]
J. Pino, L. Puzon, J. Gu, X. Ma, A. D. McCarthy, and **D. Gopinath**
International Conference on Spoken Language Translation (IWSLT), 2019.

[14] "Deep Multimodal Fusion for Persuasiveness Prediction" [pdf]
B. Nojavanasghari*, **D. Gopinath***, J. Koushik*, T. Baltrušaitis, and L. P. Morency
ACM International Conference on Multimodal Interaction (ICMI), 2016.